AMAN RAJ

Arizona State University
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EDUCATION

PhD student, Civil, Environmental & Sustainable Engineering 2024-
Arizona State University

MS Sustainable Systems 2022-2024
Rochester Institute of Technology

MS Regulatory Governance 2018 – 2020
Tata Institute of Social Sciences, Mumbai, India

B. Tech Electronics & Communication 2012-2016
Vellore Institute of Technology, Vellore, India

RESEARCH EXPERIENCE

Arizona State University Tempe, AZ
Graduate Research Assistant (with Dr. Dwarakanath Ravikumar) Aug 2024 – present

- Led a research project in collaboration with the industry to publish the first US-specific life cycle assessment of recycling Cadmium Telluride solar photovoltaic.
- Developed a supply chain optimization model integrating life-cycle assessment with geospatial truck-route optimization to evaluate 1,576 US CdTe PV frame pathways, identifying secondary aluminum as the lowest-CO_2 option and quantifying transport break-even distances that shift material preference.
- Developing a rapid, non-contact, non-destructive PV module health diagnostic using hyperspectral imaging and machine learning to support solar PV circularity

Rochester Institute of Technology Rochester, NY
Graduate Research Assistant (with Dr. Callie Babbitt) Jan 2023- Aug 2024

- Assisted in a US DOE-supported project that assesses the Life Cycle Impact of the remanufacturing process for reusing the Lithium-ion battery pack (LIB) in a Second Life Battery Application.

PUBLICATION

Journal papers

Raj, A., Ravikumar D., Winicov, M., "*Anticipatory Life Cycle Assessment of an Industry-scale CdTe Photovoltaic Recycling Process in the United States*" (Published), Resource, Conservation and Recycling, March 2026

Industry Reports

Shukla, S. Sen, T. and **Raj, A** (2020). "*Impact of E-mobility Transitions on Job: A Case Study of Jaipur, Rajasthan*." [Online] Cuts-ccier.org. Available here

Shukla, S. Sen, T. Tiwari, A. and **Raj, A** (2022). *Whitepaper on Regional Regional Parliamentary Forum on Energy Cooperation & Energy Trade in South Asia."* [Online] Irade.org. Available here

Policy Briefs

Grover, S. and **Raj. A.** (December 2020). "*Power Sector Transformation: Enabling Civil Society Participation for Inclusive Reforms."* [Online] Cuts-ccier.org. Available here

Raj, A. (June 2020). *Ease of E-Mobility in India: Policy Imperatives of COVID-19 on 2W, 3W, and 4W Electric Vehicles."* [Online] Cuts-ccier.org. Available here

Advocacy Papers

Raj, A. Shukla, S. and Tiwari, A. (December 2020). "*Response Paper: CUTS Comments on Draft RERC (Consumer Grievance Redressal Forum, Electricity Ombudsman and Consumer Advocacy) Regulations, 2020."* [Online] Cuts-ccier.org. Available here

Shukla, S. **Raj, A**. and Saroha, A. (July 2020). "*Bill Blow up: Draft Electricity Act (Amendment) Bill, 2020."* [Online] Cuts-ccier.org. Available here

Shukla, S. Sen, T. and **Raj, A**. (June 2020). "*Response Paper: CUTS Comments on RERC s Suo-moto Draft Order (Charging Infrastructure, Tariff and Other Regulatory Issues for Electric Vehicles)."* [Online] Cuts-ccier.org. Available here

Shukla, S. **Raj, A**. and Saroha, A. (June 2020). "*Response Paper: CUTS comments on Draft Electricity Act (Amendment) Bill 2020."* [Online] Cuts-ccier.org. Available here

Media Articles

Raj, A. (January 2021). "*Skill Development: Key to Atmanirbhar in Electric Mobility Transition.* Clean Mobility Shift platform." Available here

Shukla, S. and **Raj, A**. (September 2020). "*Indian Electric Vehicle Conundrum: A tale of opportunities amid uncertainties."* ET Energy. Available here

PRESENTATIONS

- "Research by Grads", Circular Arizona (City of Phoenix), Tempe, AZ, April 2026 [*oral*]
- 16th Annual School of Sustainable Engineering and the Built Environment Graduate Poster Symposium at Arizona State University, Tempe, AZ, Feb 2026 [*Poster*]
- 2025 American Institute of Chemical Engineers (AIChE) Annual Meeting, Boston, MA, Nov 2025 [*Oral*]
- 32nd International Symposium on Sustainable Systems and Technology (ISSST) Conference, Minneapolis, MN, June 2025 [*Oral & Poster*]

MENTORSHIP & OUTREACH

- ISSST Conference: Member of the planning committee [2025-present]
- Undergraduate students (ASU): Kendall Fender

- Master student (ASU): Pranava krishnaprasad, Anamika Ghopade

INDUSTRY EXPERIENCE

COI Energy Brooklyn, NY (Remote)
Summer Intern June 23 - Aug 23

- Led and spearheaded the Community Battery Storage project inception and development. My role encompassed creating a consumer optimization model to develop pricing models that assess the economic feasibility of energy projects.

Consumer Unity & Trust Society (CUTS) Jaipur, India
Research Associate Apr 2020 – Feb 2021

- Year-long researcher in a not-for-profit organization that works in the field of Public Policy, particularly Energy Access, Electric Mobility, and Just Transitions in the Energy Sector.

GENWORKS Healthcare Bengaluru, India
Project Management Executive Jun 2016 – Feb 2017

- New initiatives marketing head in the Bihar and Jharkhand region of India.

AWARDS & HONORS

- SSEBE Travel Award, Arizona State University 2025
- Fulton Fellowship, Arizona State University 2024-25
- Graduate Merit Scholarship, Rochester Institute of Technology 2022